

Do Persistence-Diagram Distances Track Structural Fidelity in Wind-Field Super-Resolution?

Phase B: Patchwise Persistence Diagrams and Wasserstein Metrics

Arjun Dadhwal

January 2026

Abstract

Super-resolution (SR) is commonly evaluated using pixel-based metrics such as PSNR and MSE. While useful for measuring average intensity error, these metrics can miss structural distortions that are visually and physically meaningful in scientific fields. In Phase A, I showed that Euler characteristic curves (a lightweight topological descriptor) reveal differences between bicubic interpolation, CNN-based SR, and GAN-based SR on two-dimensional wind fields.

In Phase B, I move from Euler curves to *persistence diagrams* (PDs) and evaluate SR outputs using Wasserstein distances between PDs computed on wind-speed patches. I compute PD_0 (connected components) and PD_1 (holes/loops) and compare bicubic, CNN_MRHR, and GAN_MRHR. Across the evaluated samples, GAN_MRHR achieves the best PD_1 fidelity (lower W_1 , and lower W_2 on a compute-limited audit subset) despite lower PSNR; bicubic achieves the highest PSNR yet the worst topological distances. These results strengthen the claim that topological distances capture complementary aspects of fidelity that PSNR alone does not.

1 Introduction

Scientific super-resolution differs from natural-image SR in an important way: the goal is not merely perceptual sharpness, but preservation of meaningful structures (e.g., coherent flow features, fronts, vortices). Pixel metrics such as PSNR summarize average error and can reward overly smooth predictions that erase small-scale structure. This motivates topology-aware evaluation: topological summaries are designed to be insensitive to small perturbations, yet sensitive to connectivity changes.

Phase A demonstrated that Euler characteristic curves separate methods in wind-field SR and can reveal a PSNR–topology mismatch. Phase B makes the topological signal more granular by using persistence diagrams (PDs). PDs isolate individual topological events

(birth/death of components and holes) and support metric comparisons via Wasserstein distances.

Phase B question. Do PD distances (Wasserstein) differentiate SR methods, and do they highlight cases where GAN SR preserves higher-order structure (PD_1) even when PSNR is worse?

2 Background

2.1 Scalarization of the wind field

Each wind sample is a 2D vector field $\mathbf{w}(x) = (u(x), v(x))$ on a grid. Persistence diagrams are defined on scalar functions, so I evaluate topology on the derived wind speed

$$s(x) = \|\mathbf{w}(x)\|_2 = \sqrt{u(x)^2 + v(x)^2}.$$

Speed is physically meaningful and commonly used as a scalar proxy for flow intensity. (A limitation is that scalarization discards directional information; I return to this in Sec. 7.)

2.2 Persistence diagrams on a cubical complex

Given a scalar field f on a grid, a *filtration* sweeps a threshold τ and tracks the topology of the induced sublevel (or superlevel) sets. In 2D, the two most relevant homology dimensions are:

- PD_0 : connected components (0D homology),
- PD_1 : holes/loops (1D homology).

A persistence diagram is a multiset of points (b_i, d_i) where b_i is the threshold where a feature appears and d_i is when it disappears. Points near the diagonal ($d_i \approx b_i$) have low persistence and are often treated as noise.

Superlevel vs sublevel. For nonnegative speed s , superlevel persistence on s is equivalent to sublevel persistence on $-s$. The evaluation code uses this standard conversion by normalizing s to $[0, 1]$ and computing persistence on $-s$.

2.3 Wasserstein distances between persistence diagrams

To compare an SR prediction $\hat{s}$ to ground truth s^{GT} , I compute distances between their persistence diagrams. Intuitively, a Wasserstein distance measures the minimum “transportation cost” of matching points between the two diagrams (with the option to match to the diagonal, representing feature deletion).

60 For $p \in \{1, 2\}$, the p -Wasserstein distance is:

$$W_p(D_1, D_2) = \left(\inf_{\gamma} \sum_{x \in D_1} \|x - \gamma(x)\|_{\infty}^p \right)^{1/p},$$

61 where γ ranges over matchings between points in D_1 and $D_2 \cup \Delta$ and Δ is the diagonal.

62 Lower is better (more topological agreement). In practice:

- 63 • W1 is typically more robust and easier to estimate,
- 64 • W2 penalizes large mismatches more strongly but can be substantially more expensive.

66 3 Experimental Setup

67 3.1 Data and SR methods

68 I evaluate medium-resolution to high-resolution (MR→HR) wind SR using:

- 69 1. **Bicubic** interpolation baseline,
- 70 2. **CNN_MRHR** (pixel/MSE-oriented SR),
- 71 3. **GAN_MRHR** (adversarial SR aiming to preserve sharper structure).

72 **Data availability.** The provided TFRecords contain 5
73 records each (example_data/wind_LR-MR.tfrecord and
74 example_data/wind_MR-HR.tfrecord), i.e., 10 total records across the two
75 datasets. For Phase B, the MR→HR dataset is used. Because this is a small sample size,
76 results should be interpreted as preliminary trends.

77 3.2 Patchwise persistence evaluation

78 Computing PDs on full fields is expensive and may blur local structure. I therefore evaluate
79 on patches of the speed field:

- 80 • patch size: 160×160 ,
- 81 • stride: 160 (non-overlapping),
- 82 • 9 patches per sample $\Rightarrow$ 45 patches per run for 5 samples.

83 For each patch, I compute:

- 84 • PSNR between SR and GT on the speed patch,
- 85 • W1 distance between PD_0 diagrams (SR vs GT) and between PD_1 diagrams,
- 86 • W2 distances on a compute-limited audit subset (below).

87 3.3 Diagram pruning and compute constraints

88 To stabilize PD distances and reduce compute, the evaluation prunes diagrams:

- 89 • minimum persistence threshold: `min_pers=0.01`,
- 90 • maximum points per diagram: `max_pts=120`.

91 **W2 audit subset.** W2 is more expensive than W1. On compute-limited hardware, I com-
92 pute W2 only for every k -th patch (`every=3`) in a deterministic manner. W2 values should
93 be interpreted as partial evidence until full evaluation is feasible.

94 4 Reproducibility and Replication Steps

95 This section gives a complete, file-accurate recipe for reproducing the Phase B results (CSV,
96 plots, and report). Paths are relative to the project root (the repository shown in the directory
97 tree).

98 4.1 Inputs expected on disk

99 **Paired inference dumps (required).** Phase B assumes you already have paired MR→HR
100 inference outputs for both CNN and GAN in:

- 101 • `data_out/wind_mrhr_cnn/` containing `dataGT.npy`, `dataIN.npy`,
102 `dataSR.npy`, `idx.npy`
- 103 • `data_out/wind_mrhr_gan/` containing the same four files

104 These can be generated using the inference scripts from Phase A:

```
105 python run_paired_wind_mr_hr.py # GAN MR->HR inference  
106  
107 python run_paired_wind_mr_hr_cnn.py # CNN MR->HR inference  
108
```

109 **Important:** downstream scripts assume the folder names above. If PhIREGANs writes
110 to timestamped directories under `data_out/`, copy/rename the output folders to match
111 `wind_mrhr_cnn` and `wind_mrhr_gan`.

112 Phase B scripts (required).

- 113 • `phase_b_persistence_eval.py`: computes per-patch PSNR and PD dis-
114 tances and writes `phase_b_persistence_results.csv`.
- 115 • `plot_phase_b_summary.py`: reads the CSV and writes bar plots, scatter plots,
116 and `phase_b_summary_stats_speed.csv`.
- 117 • `make_phase_b_summary.py`: generates `figs_phase_b/phase_b_results_brief.pdf`
118 (a compact PDF report).

119 4.2 Reproduce using the provided shell script

120 The repository includes `reproduce_phase_b.sh`, which builds a Docker image and
121 runs all Phase B steps. From the project root:

```
122 bash reproduce_phase_b.sh
```

125 **Expected outputs.** After a successful run:

- 126 • raw patchwise metrics: `phase_b_persistence_results.csv`
- 127 • plots and summaries: `figs_phase_b/` containing:
 - 128 – bar plots: `bar_mean_psnr_speed.png`,
129 `bar_mean_w1_pd0_speed.png`, `bar_mean_w1_pd1_speed.png`,
130 `bar_mean_w2_pd0_speed.png`, `bar_mean_w2_pd1_speed.png`
 - 131 – scatter plots: `scatter_psnr_w1_pd0_speed_CNN_MRHR.png`,
132 `scatter_psnr_w1_pd0_speed_GAN_MRHR.png`,
133 `scatter_psnr_w1_pd1_speed_CNN_MRHR.png`,
134 `scatter_psnr_w1_pd1_speed_GAN_MRHR.png`
 - 135 – summary table as CSV: `phase_b_summary_stats_speed.csv`
 - 136 – brief PDF report: `phase_b_results_brief.pdf`

137 4.3 Manual reproduction: the exact Docker command

138 For macOS Apple Silicon, the topo image is run with `-platform linux/amd64`. The
139 commands below match the repository script:

```
140 docker build -f Dockerfile.topo -t phire-topo:phaseb .
141
142
143 docker run -it --rm \
144     --platform linux/amd64 \
145     -u "$(id_u)": "$(id_g)" \
146     -e HOME=/work \
147     -v "$PWD":/work \
148     -w /work \
149     phire-topo:phaseb \
150     bash -lc '
151     ____set_e
152     ____python_phase_b_persistence_eval.py
153
154     ____mkdir_p_/work/figs_phase_b
155     ____PHIRE_OUTDIR=/work/figs_phase_b_python_plot_phase_b_summary.py
156
157     ____python_make_phase_b_summary.py
```

Table 1: Phase B summary on wind **speed**. Each row aggregates 5 samples (45 patches) per method. W2 is computed only on an audit subset (compute-limited).

Method	PSNR $\uparrow$	W1(PD0) $\downarrow$	W1(PD1) $\downarrow$	W2(PD0) [†] $\downarrow$	W2(PD1) [†] $\downarrow$	W2 count
BICUBIC	27.51 $\pm$ 3.11	6.65 $\pm$ 1.55	7.52 $\pm$ 2.88	0.702 $\pm$ 0.125	–	12/15
CNN_MRHR	24.42 $\pm$ 2.68	5.22 $\pm$ 1.02	6.25 $\pm$ 2.12	0.604 $\pm$ 0.131	–	12/15
GAN_MRHR	22.20 $\pm$ 1.62	5.30 $\pm$ 1.46	4.60 $\pm$ 1.04	0.520 $\pm$ 0.114	0.478 $\pm$ 0.072	13/16

[†]W2 is computed only on the deterministic audit subset (every=3). Missing W2 entries can occur if a sample has no audited patches or if a diagram is empty after pruning.

158
159

‘

160 4.4 Sanity check: verify TFRecord counts

161 To confirm the number of records in the provided example datasets:

```

162 python - <<'PY'
163 import tensorflow as tf
164 def count(path):
165     c = 0
166     for _ in tf.data.TFRecordDataset([path]):
167         c += 1
168     return c
169 print("wind_LR-MR:", count("example_data/wind_LR-MR.tfrecord"))
170 print("wind_MR-HR:", count("example_data/wind_MR-HR.tfrecord"))
171 PY
172
173
```

174 5 Results

175 5.1 Summary table (per-sample aggregation)

176 The raw CSV (phase_b_persistence_results.csv) contains per-patch metrics.

177 For reporting, I aggregate in two stages:

- 178 1. average patch metrics within each sample,
- 179 2. compute mean $\pm$ std across samples (5 samples per method).

180 Table 1 reports this per-sample aggregation (higher PSNR is better; lower W1/W2 is
181 better). W2 is computed on an audit subset; the final column reports the number of finite
182 W2 values observed for PD₀ and PD₁ across all audited patch instances.

183 5.2 Bar plots

184 Figure 1 shows three clear patterns.

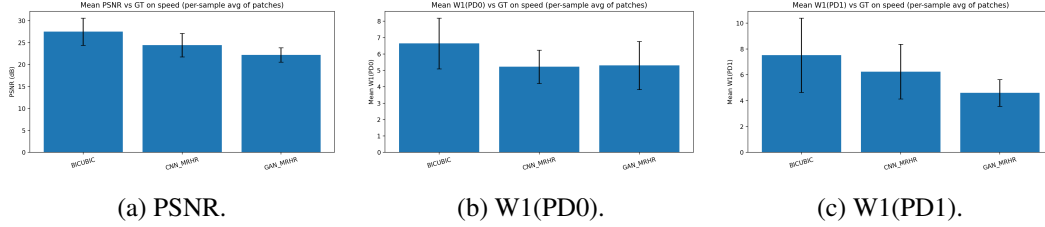


Figure 1: Phase B (speed): mean $\pm$ std across samples after averaging patches within each sample.

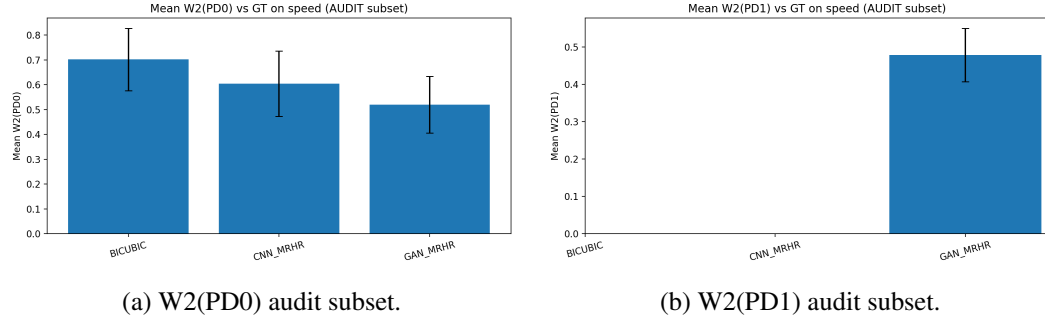


Figure 2: W2 distances on the audit subset. Treat as partial evidence (compute-limited).

- **PSNR is not aligned with topological fidelity.** Bicubic achieves the highest PSNR, yet it is worst under both W1(PD0) and W1(PD1). This reproduces the Phase A theme: a method can be “good” under a pixel metric while being structurally worse.
- **GAN improves PD₁ substantially.** GAN_MRHR yields the lowest W1(PD1) among learned methods and is much lower than bicubic. This suggests that GAN SR better preserves loop/hole structure in wind-speed patches.
- **PD₀ is mixed.** CNN_MRHR and GAN_MRHR are close on W1(PD0), consistent with PD₀ being influenced by many small extrema and components; smoothing (CNN) and texture (GAN) can create different but similarly sized PD₀ errors.

Figure 2 supports the same qualitative story: GAN_MRHR is best on W2(PD0) and also yields a finite W2(PD1) estimate on the audited subset. Because W2 is computed only on a subset, these results should be validated at larger scale once compute allows.

5.3 Scatter plots: PSNR vs topology per patch

The scatter plots in Fig. 3 are useful for diagnosing whether a PSNR–topology “tradeoff” is inevitable. Two observations stand out:

- **The relationship is only partially monotone.** In both CNN and GAN, higher PSNR patches tend to have lower W1, but there are clear outliers: patches with moderate PSNR yet high topological error. This is precisely where topology adds diagnostic power.

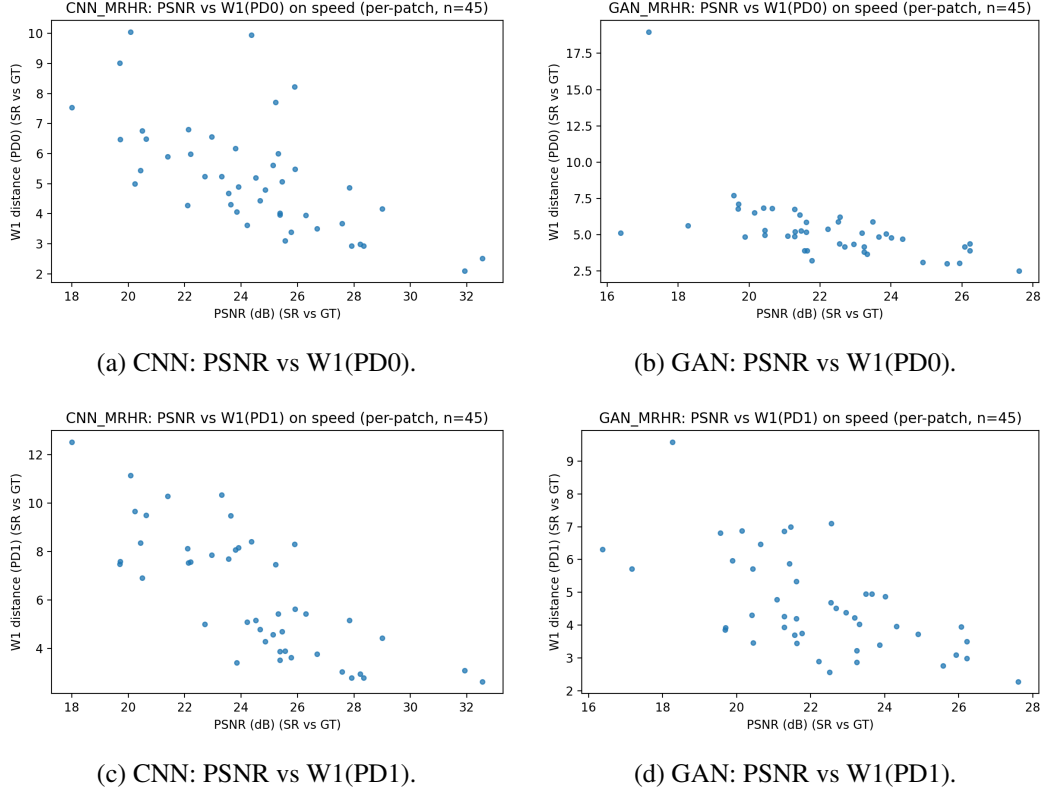


Figure 3: Per-patch PSNR vs topology distance (speed). Lower W1 is better.

- **Some patches achieve both good PSNR and good topology.** The existence of points in the “upper-right” PSNR region with relatively low W1 argues against a strict trade-off. It suggests that with better training or loss design, it may be possible to improve both pixel and topological fidelity.

6 Discussion

6.1 Promising signals

The results are “promising but mixed,” and can be summarized cleanly:

- **GAN shows topology-aware potential (PD₁).** GAN_MRHR has substantially lower W1(PD1) than CNN_MRHR while also showing favorable W2(PD0) and a finite W2(PD1) estimate on the audited subset. This supports the hypothesis that GAN SR can better preserve higher-order structure in speed fields.
- **Topology separates methods beyond PSNR.** Bicubic ranking flips between PSNR and PD distances: it wins PSNR but loses topology. This is strong evidence that PD distances capture information PSNR does not.

6.2 Ambiguities and concerns

- **PSNR–topology mismatch.** GAN_MRHR has the lowest PSNR. This can be framed either as a genuine tradeoff (GAN introduces texture that increases pixel error) or as PSNR being misaligned with the science-relevant structures. The scatter plots suggest the mismatch is not absolute.
- **PD₀ is inconclusive.** GAN and CNN are close on W1(PD₀). A plausible explanation is that PD₀ is dominated by many small extrema/components; smoothing (CNN) can remove tiny components, while GAN can introduce small fluctuations, and both may incur similar costs in component matching.
- **Small sample size.** Only 5 MR→HR records are available in the provided TFRecord. Patch extraction increases the number of evaluated patches, but not the number of independent samples. Future experiments should use more distinct wind samples.
- **Compute-limited W2.** W2 is computed only on an audit subset. Once more compute is available, W2 should be computed for all patches to confirm the trends.

7 Limitations and Next Steps

Vector-field topology. This phase computes topology on speed, a scalarization of the vector field. Speed is meaningful, but it discards directionality and may hide vector-specific structures. Future work should explore topology on derived scalar diagnostics (e.g., vorticity magnitude) and/or vector-field topology descriptors.

Toward merge trees. Persistence diagrams summarize births/deaths but can lose hierarchical nesting information. Wind magnitude fields often exhibit multi-scale nested structure (fronts and vortices). Merge trees explicitly represent how components merge across thresholds and may provide a more interpretable, hierarchical assessment of structural fidelity.

Data scaling. The immediate bottleneck is data: the provided TFRecords include only 5 records per wind dataset. The next step is to acquire or generate a larger wind dataset, then re-run the SR inference pipeline to produce `data_out/...` dumps, and finally re-run the Phase B evaluation unchanged.

8 Conclusion

Phase B demonstrates that persistence-diagram distances provide a richer topology-aware evaluation of wind-field SR than pixel metrics alone. In this preliminary study, GAN_MRHR improves PD₁ Wasserstein distances relative to CNN_MRHR despite lower PSNR, while bicubic highlights a sharp decoupling between PSNR and structural fidelity.

250 These results motivate scaling the dataset and moving to hierarchical descriptors such as
251 merge trees to assess multi-scale flow structure preservation.

252 **References**

- 253 [1] H. Edelsbrunner and J. Harer. *Computational Topology: An Introduction*. American
254 Mathematical Society, 2010.
- 255 [2] The GUDHI Project. *GUDHI User and Reference Manual*.
- 256 [3] S. Oudot. *Persistence Theory: From Quiver Representations to Data Analysis*. AMS,
257 2015.